Categorical Data

Selected lecture slides

SOURCE

2.2 - Considering Categorical Data
Mine Çetinkaya-Rundel
Curry Hilton (Google Slides conversion)

https://docs.google.com/presentation/d/1D8PHarhuhj_FLDc2b7l7BoQ3euaHN0dv0V0T5-qttE4

Selected original PDF pages: 3-5, 7, 11-14, 16.

Added this source notice; retained complete teaching pages. Removed staged animation duplicates, an externally sourced pie-chart illustration and resource advertisements. Source page 1 identifies the creators and links to the license below.

CC-BY-SA-3.0-US
https://creativecommons.org/licenses/by-sa/3.0/us/

Prepared 2026-10-17. This selection is not an official OpenIntro product.
Original authors retain credit. Page mapping is included in manifest.json.

Contingency Tables

A table that summarizes data for two categorical variables is called a *contingency table*.

Contingency Tables

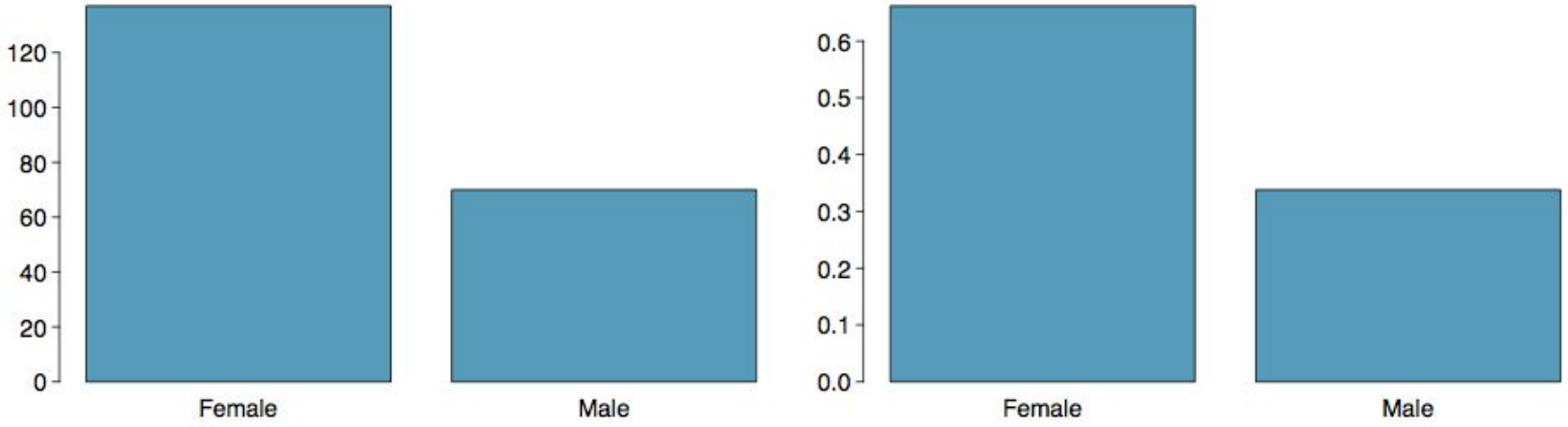
A table that summarizes data for two categorical variables is called a *contingency table*.

The contingency table below shows the distribution of students' genders and whether or not they are looking for a spouse while in college.

		looking for spouse		
		No	Yes	Total
gender	Female	86	51	137
	Male	52	18	70
	Total	138	69	207

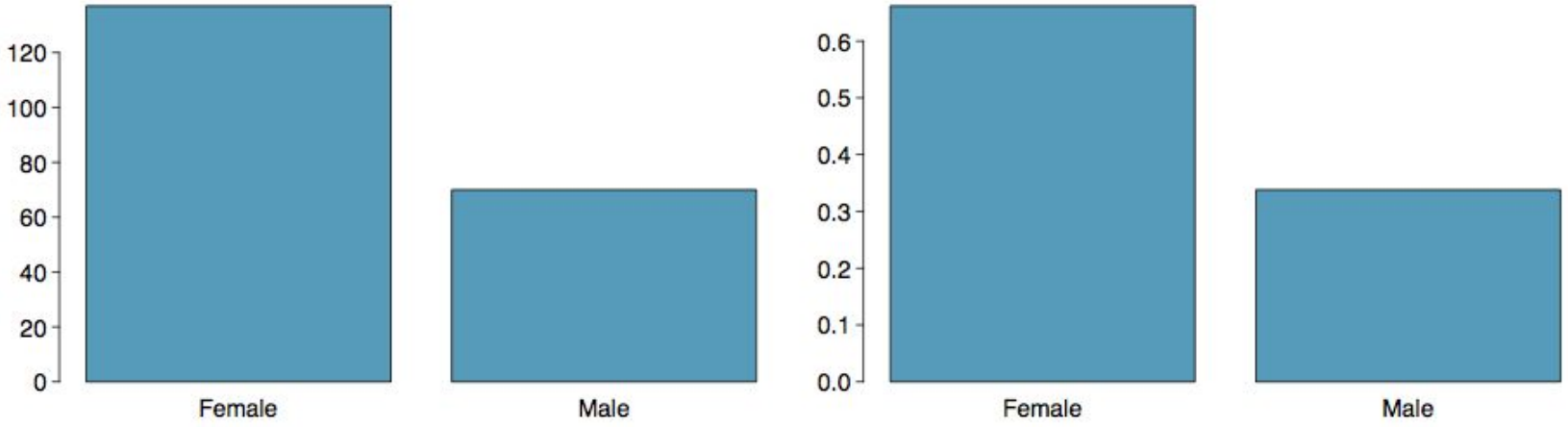
Bar Plots

A *bar plot* is a common way to display a single categorical variable. A bar plot where proportions instead of frequencies are shown is called a *relative frequency bar plot*.



Bar Plots

A *bar plot* is a common way to display a single categorical variable. A bar plot where proportions instead of frequencies are shown is called a *relative frequency bar plot*.



How are bar plots different than histograms?

Bar plots are used for displaying distributions of categorical variables, while histograms are used for numerical variables. The x-axis in a histogram is a number line, hence the order of the bars cannot be changed, while in a bar plot the categories can be listed in any order (though some orderings make more sense than others, especially for ordinal variables.)

Choosing the Appropriate Proportion

Does there appear to be a relationship between gender and whether the student is looking for a spouse in college?

		looking for spouse		
		No	Yes	Total
gender	Female	86	51	137
	Male	52	18	70
	Total	138	69	207

To answer this question we examine the row proportions:

- % Females looking for a spouse: 51 / 137 ~ 0.37
- % Males looking for a spouse: 18 / 70 ~ 0.26

Bar plots with two variables

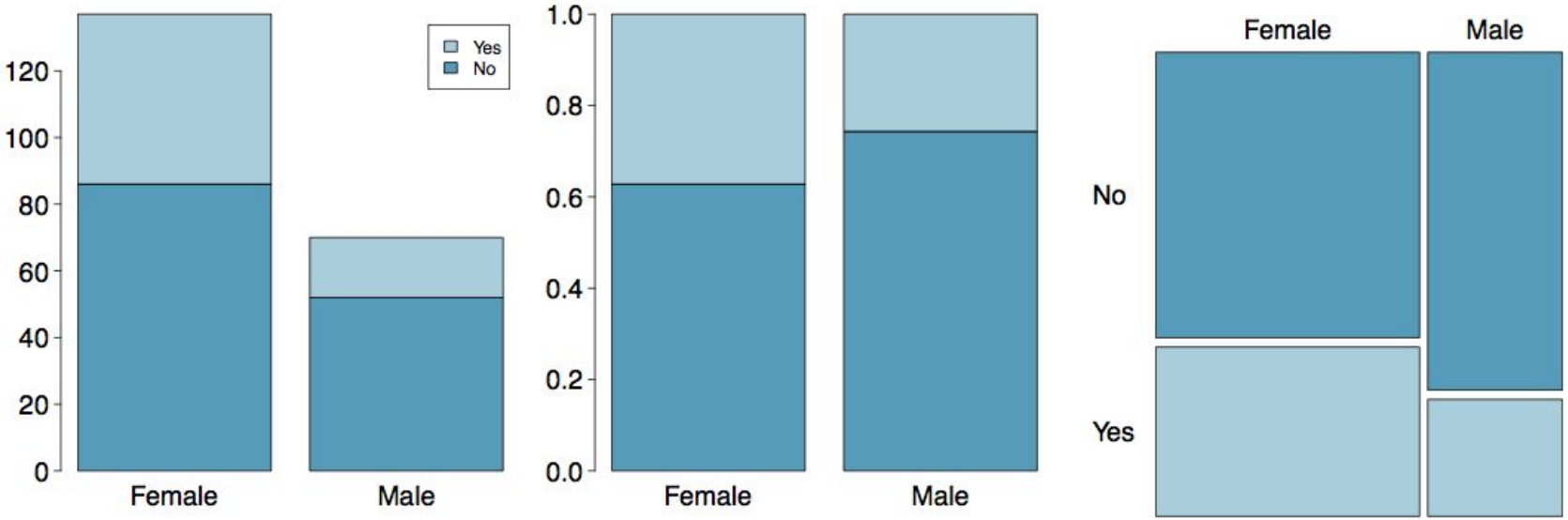
- *Stacked bar plot:* Graphical display of contingency table information, for counts.

- *Side-by-side bar plot:* Displays the same information by placing bars next to, instead of on top of, each other.

- *Standardized stacked bar plot:* Graphical display of contingency table information, for proportions.

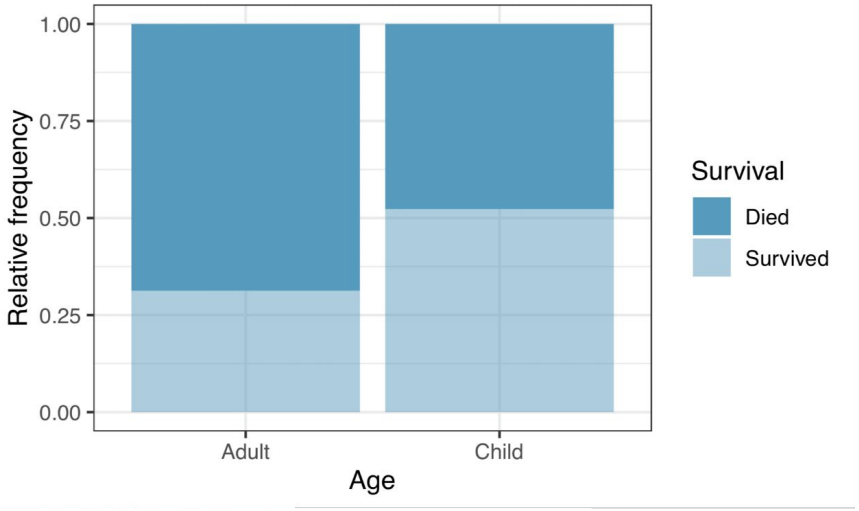
Segmented Bar and Mosaic Plots

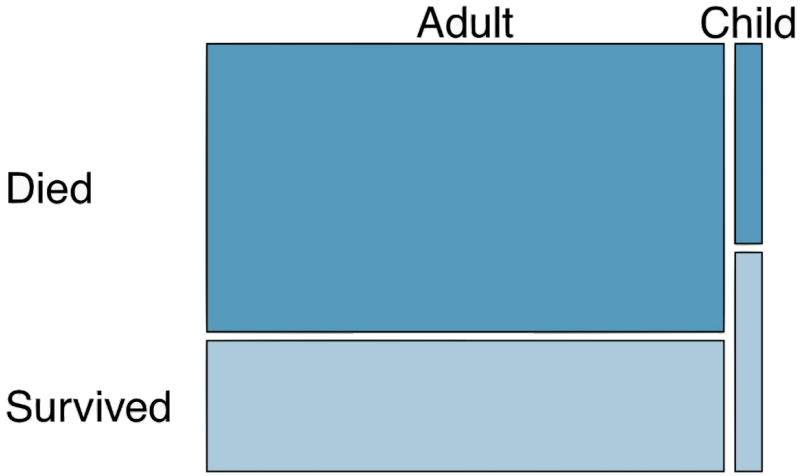
What are the differences between the three visualizations shown below?



Mosaic plots

What are the differences between the two visualizations shown below?





Comparing Numerical Data Across Groups

Does there appear to be a relationship between class year and number of clubs students are in?

